

Mohak Rathod

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Education

Arizona State University

Aug 2025–May 2027

M.S. Computer Science

GPA: 3.56

Relevant Coursework: Cloud Computing, Forensics Computing, Planning and Learning Methods of AI

University of Mumbai

Jun 2025

B.Tech. Computer Science and Engineering (IoT, Cybersecurity & Blockchain)

CGPA: 9.04/10

Technical Skills

Languages: Python, Go, Rust, C++, Java, TypeScript

AI/ML: PyTorch, Scikit-learn, LangGraph, RAG, MCP, Ollama

Backend: FastAPI, gRPC, REST APIs, Docker, Kubernetes

Databases: Neo4j, PostgreSQL, Redis, ChromaDB

Cloud DevOps: AWS, Git, GitHub Actions, Prometheus, Grafana **Security:** MITRE ATT&CK, CVE, EPSS, Gitleaks, Trivy, Bandit

Research

MOSAIC — Multi-Agent AI Coaching System, ASU VISA Lab

May 2026–Present

Supervisor: Prof. Ming Zhao

- Developing a six-agent coaching architecture integrating behavioral sensing with a GROW-based coaching workflow using AgentScope.

Malicious URL Classification (MMAUC)

Aug 2024–May 2025

- Developed an ensemble learning pipeline with GAN-based augmentation, improving class balance by 22%, achieving 93% accuracy and reducing false positives by 18%. Under peer review.

Projects

ARGUS — Multi-Agent Security Investigation System

Jun 2026– Jul 2026

Python · LangGraph · Claude · Neo4j · MCP · FastAPI · Streamlit

- Built a multi-agent system that autonomously investigates security alerts by correlating CVE/EPSS intelligence, Neo4j attack graphs, and historical exposure into MITRE ATT&CK-aligned incident reports.
- Implemented deterministic severity scoring through rule-based aggregation instead of LLM judgment, reducing prompt-injection attack surface.
- Evaluated on 30 real, synthetic, and adversarial alerts, achieving 100% task completion with no observed hallucinations or prompt-injection hijacks across seven attack vectors.

VulnGraph — AI-Powered Application Security Platform

Dec 2025– May 2026

Python · Neo4j · ChromaDB · BM25 · FastAPI · Ollama · MCP

- Built an ASPM platform integrating Gitleaks, Trivy, and Bandit into Neo4j attack graphs.
- Developed a hybrid RAG pipeline over 120 security documents for vulnerability-specific retrieval.
- Implemented an automated remediation pipeline and benchmarked multiple local LLMs using relevance, faithfulness, and CWE accuracy.

Distributed Systems & Cloud Projects

2025–2026

Go, Kubernetes, Redis, AWS, PyTorch

- Built a distributed token-bucket rate limiter sustaining 3,000 requests/s with 1.57 ms average latency.
- Built a 10-node federated learning cluster on AWS and Raspberry Pi using IoT Greengrass, achieving 97.7% MNIST accuracy.
- Designed a custom AWS autoscaler supporting 0–15 EC2 instances with 0.11 s scale-in time.

Publications

- **Blockchain Powered File Sharing System.** DOI: 10.53555/AJBR.v27i6S.5138

- **Votix: Empowering Democratic Processes Through Decentralized and Biometric-Driven Voting Systems.** Springer, *Reliability and Safety Assessment for Safety Critical Systems: Select Proceedings of NCRS 2025*. DOI: 10.1007/978-981-95-6140-7

Experience

SOC Analyst Intern — Syntel Pvt. Ltd. (Atos/Eviden Group)

Summer 2023, 2024

- Monitored enterprise security events using IBM QRadar, ArcSight, LogRhythm, and Zscaler while supporting incident response and Power BI reporting.